

MORTEZA KHOSHSIMAIE CHENAR

Ph.D. in Irrigation and Drainage Engineering

Agricultural AI | Machine Learning | Climate-Smart Agriculture | Decision Support Systems

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PROFILE SUMMARY

Agricultural AI researcher with expertise in climate-smart agriculture, decision support systems, and machine learning applications for water and nutrient management. My research combines data-driven modeling, environmental analytics, crop modeling, and software development to improve agricultural productivity, resource-use efficiency, and climate resilience. I have developed an AI-powered web-based decision support system for greenhouse management and published research on machine learning, irrigation optimization, evapotranspiration modeling, soil salinity, and precision water management. My long-term research goal is to develop intelligent digital solutions that support sustainable and climate-resilient agricultural systems.

RESEARCH EXPERTISE

- Agricultural Artificial Intelligence and Machine Learning
- Decision Support Systems for Smart Agriculture
- Climate-Smart Agriculture
- Environmental Data Analytics
- Precision Irrigation and Fertigation
- Crop and Agro-hydrological Modeling
- Water and Nutrient Use Efficiency
- Remote Sensing and Environmental Monitoring

EDUCATION

University of Tehran, Iran | **Ph.D. in Irrigation and Drainage Engineering: Sep 2019 – Feb 2026**

- GPA: 17.44 / 20.00
- Thesis: Development of a Decision Support System for Scheduling Irrigation and Fertilization of Greenhouse Cucumber in Conditions of Using Waters with Different Quality.
 - Developed an AI-powered web-based decision support system for greenhouse irrigation and fertigation.
 - Integrated machine learning models with environmental and crop data.
 - Evaluated irrigation strategies under different water quality conditions.

University of Tehran, Iran | M.Sc. in Irrigation and Drainage Engineering: *Sep 2016 – Sep 2018*

- GPA: 16.75 / 20.00
- Thesis: The Effect of Water Salinity on the Growth of Different Varieties of Maize.

Payame Noor University, Ajab Shir, Iran | B.Sc. in Water and Soil Engineering: *Sep 2011 – July 2015*

- GPA: 17.17 / 20.00
- Achievement: Ranked 1st among B.Sc. students of the 2011 entrance class.

SOFTWARE DEVELOPMENT & PROJECTS

GreenSmart-DSS: AI-based Decision Support System for Smart Greenhouse Management

- Developed a web-based Decision Support System (DSS) for irrigation and fertigation scheduling of greenhouse cucumber under variable water quality conditions.
- Integrated machine learning models for environmental and agricultural predictions (including evapotranspiration and radiation-related components).
- Designed system modules for ET_o and ET_c estimation, crop irrigation requirement, fertilization optimization, solar radiation estimation, climate feasibility analysis (greenhouse energy demand analysis).
- Implemented data-driven decision-making workflows using real-time and historical climate datasets.
- Built backend system using Django and Python, enabling scalable integration of ML models with agricultural databases.
- Applied optimization and statistical learning methods to support sustainable water and nutrient management strategies.

Technologies: Python, Django, HTML/CSS, Scikit-learn, Pandas, Numpy, Optuna

Root-Zone Soil Moisture Monitoring Using Remote Sensing and Simulation Modeling

- Developed and integrated framework combining the SWAP agro-hydrological model with multi-source satellite data (Sentinel-2, Landsat-8) to estimate daily root-zone soil moisture at high spatial resolution for precision irrigation management
- Applied inverse modeling and data assimilation techniques using genetic algorithms to optimize soil hydraulic parameters and incorporate vegetation indices, reducing dependence on in-situ measurements.
- Validated the approaches across multiple agricultural fields (wheat and maize) under diverse climatic and soil conditions in Iran.
- Demonstrated applicability for variable-rate irrigation strategies to improve water use efficiency in water-scarce agricultural regions
- Utilized advanced remote sensing methods including OPTRAM and optical satellite image processing for soil moisture and vegetation monitoring.

Technologies: SWAP, MATLAB, Remote Sensing (Sentinel-2, Landsat-8), Genetic Algorithms.

PUBLICATIONS

Journal Papers (International Journals)

1. Noory, H., **Khoshshima, M.**, Tsunekawa, A., Tsubo, M., Haregeweyn, N., & Pashapour, S. (2025). Developing a method for root-zone soil moisture monitoring at the field scale using remote sensing and simulation modeling. *Agricultural Water Management*, 308, 109263.
2. **Khoshsimaie Chenar, M.**, Noory, H., Soltani Salehabadi, F., & Motesharezade, B. (2026). Salinity tolerance threshold in greenhouse cucumber cultivation: a comparative analysis of mathematical models. *Irrigation Science*, 44(2), 35.

Journal Papers (Persian with English Abstract)

3. Hoseini, S. M., & **Khoshsimaie Chenar, M.** (2025). Application of machine learning algorithms in groundwater level prediction in the Ardabil aquifer. *Iranian Journal of Soil and Water Research*, 56(4), 1041-1057.
4. **Khoshsimaie Chenar, M.**, Noory, H., Liaghat, A., Soltani Salehabadi, F. and Motesharezadeh, B. (2025). Evaluation of the accuracy of different machine learning algorithms in predicting greenhouse cucumber crop evapotranspiration. *Water and Irrigation Management*, 15(3), 563-583.
5. **Khoshsimaie Chenar, M.**, Tafteh, A., & Ebrahimipak, N. (2025). Modeling Greenhouse Cucumber Evapotranspiration Using Machine Learning: A Random Forest Approach Versus Traditional and Non-linear Crop Coefficients. *Water and Soil Management and Modelling*, 5(2), 69-87.
6. **Khoshsimaie Chenar, M.**, Liaghat, A., Noory, H., Soltani Salehabadi, F., & Motesharezadeh, B. (2025). Impact of Different Salinity Levels and Irrigation Water Amounts on Yield and Water Productivity of Greenhouse Cucumber in Two Autumn-Winter and Spring-Summer Growing Periods. *Water Management in Agriculture*.
7. Golshani, H., & **Khoshsimaie Chenar, M.** (2024). Evaluation of AquaCrop and SWAP models in simulating the growth and biomass of different maize cultivars under the conditions of using saline water with drip irrigation system. *Iranian Journal of Soil and Water Research*, 55(4), 615-636.
8. **Khoshsimaie Chenar, M.**, Noory, H., & Mahmoudi Molamahmoud, Z. (2021). Evaluation of SWAP model in estimating soil water content, salinity and yield of three forage maize cultivars under saline water use conditions. *Water and Irrigation Management*, 11(3), 495-512.
9. Fakhrabadi, H., & **Khoshsimaie Chenar, M.** (2021). The Effect of Deficit Irrigation and Biochar on quantitative and qualitative characteristics of Basil. *Iranian Journal of Irrigation & Drainage*, 15(4), 941-954.
10. **Khoshshima, M.**, & Noori, H. (2019). Effect of Drip Tape Irrigation with Saline Water on Some Chemical Properties of Soil. *Journal of Water Research in Agriculture*, 33(4), 565-581.
11. **Khoshshima, M.**, & Noory, H. (2019). Effect of Irrigation Water Salinity on Yield and Agronomic Characteristics of Three Corn (*Zea mays* L.) Hybrids Using Drip-Tape Irrigation. *Iranian Journal of Soil and Water Research*, 50(8), 2037-2049.

TECHNICAL SKILLS & RESEARCH METHODS

Programming, Data Science & Computational Tools

- Python: data analysis, machine learning, scientific computing, and optimization
 - Libraries: Numpy, Pandas, Scikit-learn, Scipy, Statsmodels, optuna
 - Applied to: predictive modeling, feature engineering, model optimization, and validation

- MATLAB: numerical modeling, optimization, and agro-hydrological models calibration
- Web Development: HTML/CSS for scientific and research-based applications
- Basic experience with API integration for environmental data acquisition

Machine Learning & Agricultural Data Analytics

- Supervised machine learning for environmental and agricultural systems
 - Regression models, ensemble learning, and non-linear modeling approaches
- Model optimization and evaluation
 - Cross-validation, hyperparameters tuning, performance metrics
- Time-series and environmental data analysis
- Data visualization and interpretation
 - Matplotlib, Seaborn, Plotly

Decision Support Systems (DSS) & Software Engineering

- Design and development of AI-based Decision Support Systems for agriculture
- Full-stack development of web-based agricultural DSS platforms
- Frameworks: Django, FastAPI
- System integration of:
 - Machine learning models
 - Environmental and climate data
 - Crop simulation models
 - Irrigation and fertigation decision modules
- Experience in deploying research-based DSS applications on server/cloud environments

Agro-hydrological & Crop Modeling

- Process-based crop and soil-water modeling using:
 - SWAP, AquaCrop, HYDRUS-1D
- Applications:
 - Soil water balance and salinity dynamics
 - Crop yield and evapotranspiration simulation
 - Irrigation scheduling and water productivity analysis

Field Experiments & Environmental Data Acquisition

- Design and execution of greenhouse and field experiments
 - Irrigation and fertigation management under salinity and drought stress
- Soil, plant, and water data collection and monitoring
- Sensor-based environmental data acquisition and processing
- Experimental design, statistical analysis, and system validation

LANGUAGES

- Turkish (Azeri) – Native
 - Persian – Native
 - English – Academic Proficiency
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TEACHING & WORK EXPERIENCE

University of Tehran | Tehran, Iran *Workshop Instructor: 2021*

- Instructed M.Sc. students on the application and calibration of the SWAP Model in the Department of Irrigation and Reclamation Engineering.

Urmia Lake Restoration Program (Contractor Co.) | Ajab Shir, Iran *Irrigation Engineer: 2019 – 2020*

- Collaborated on water resource management projects aimed at the restoration of Lake Urmia.
 - Supervised field operations and monitored irrigation system efficiency.
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HONORS & AWARDS

- Member of Iran's National Elites Foundation (Bonyad Melli Nokhbegan)
 - Ranked 3rd in the National Ph.D. Entrance Exam for Irrigation and Drainage Engineering (among candidates nationwide), 2019.
 - Ranked 13th in the National M.Sc. Entrance Exam for Irrigation and Drainage Engineering, 2016.
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REFERENCES

- Dr. Abdolmajid Liaghat – Full Professor, Department of Irrigation & Reclamation Engineering, University of Tehran, Email: Aliaghat@ut.ac.ir
- Dr. Hamideh Noory – Associate Professor, Department of Irrigation & Reclamation Engineering, University of Tehran, Email: Hamideh.noory@detsi.qld.gov.au
- Dr. Forouzandeh Soltani – Assistant Professor, Department of Horticultural Science & Landscape Architecture, University of Tehran, Email: Soltanyf@ut.ac.ir
- Dr. Babak Motesharezadeh - Professor, Department of Soil Science, University of Tehran, Email: Babak.motesharezadeh@murdoch.edu.au